

PRINAN SIL

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PROFESSIONAL SUMMARY

Data and Software Engineer with more than 1.5 years of experience at Tata Consultancy Services. My day to day work includes SQL optimization, ETL automation and PL/SQL pipeline engineering for systems handling millions of records. I am well versed in Python, Oracle PL/SQL, Java and API Integrations. Additionally, I possess hands-on knowledge of batch processing, data governance, stored procedures, and report automation. I have proven experience in an ML research project, which won the ACM India Bachelor's Dissertation Award 2024. Currently seeking a Data Engineer role to apply deep SQL expertise and pipeline building experience in a data first environment.

PROFESSIONAL SKILLS

Programming: Python, Java, SQL, PL/SQL, JavaScript, C

Backend Technologies: Spring Boot, REST APIs, Microservices, JPA/Hibernate

Data Engineering / Analytics: ETL/ELT Pipelines, Data Modeling, Query Optimization, Data Analysis, Data Visualization, Feature Engineering, Data Governance, Data Orchestration

Big Data / Distributed: PySpark, Apache Spark, Databricks, Distributed Systems

Databases & Warehouses: Oracle SQL, PL/SQL, PostgreSQL, MySQL, BigQuery

Cloud / DevOps: AWS, GCP, Docker, CI/CD, Jenkins, Git, GitHub, GitHub Copilot

Tools / Platforms: Airflow, Tableau, Power BI, Jupyter, Kaggle, VS Code, Google Colab

Core Skills: System Design, Performance Tuning, API Integration, Data Pipelines, Debugging, RCA Analysis

EXPERIENCE

Tata Consultancy Services (TCS)

Systems Engineer (Nov 2024 - Present)

Oracle Database Engineering

- Developed PL/SQL batch processing pipelines for auto-rejecting deficient claims with 100% accuracy.
- Automated reporting and validation ensuring high data accuracy and operational consistency across 10M+ records.
- Implemented deductible rule logic to automate financial adjustments across millions of records with high data integrity.
- Built rule-based TDS service processing across 10,000+ monthly transactions with zero errors.
- Resolved 500+ UAT and production defects through log analysis, debugging, root cause analysis and performance tuning.

Backend and API Developer

- Built REST APIs using Spring Boot to automate claim creation, reducing manual effort and processing time by 90%.
- Delivered Short Payment feature, enabling automated secondary claims with 100% reconciliation accuracy.
- Developed a bulk approval module using optimized SQL, increasing throughput by 50% and reducing bottlenecks.
- Led the end-to-end rollout of the claims product, including schema design, rules, validation, and deployment.
- Reduced production issues by 70% through collaboration with business, QA and support teams on structural fixes.
- Integrated third-party API services for fraud detection and AI client screening, improving data governance and reliability.

Amazon ML Summer School

Student Apprentice (Sep 2023 - Nov 2023)

- Selected among 40,000+ applicants for Amazon's ML program focused on scalable ML systems and AI engineering.
- Solved ML problems using supervised, unsupervised, and deep learning techniques on real-world datasets.
- Analyzed scalable ML architectures and production level optimization strategies under mentorship of Amazon scientists.

PROJECTS / RESEARCH

Biomarker Discovery for Breast Cancer Subtype Detection *Python, Pandas, TensorFlow, Keras, Sklearn, NumPy*

- Built ML and DL models on genomic datasets for biomarker identification, achieving 90.74% classification accuracy.
- Processed large-scale genetic data using cleaning, normalization, and feature engineering, improving performance by 30%.
- Performed hyperparameter tuning and model evaluation using F1-Score and AUC-ROC metrics for accurate prediction.
- Generated data-driven insights supporting personalized treatment strategies and predictive healthcare analytics.
- Awarded ACM India Dissertation Award 2024 for research excellence in Machine Learning and Bioinformatics.

Projects on Data Analysis and Data Visualization

Tableau, Power BI, MS Excel

- Electric Vehicle Sales Analysis Using Tableau | [Link](#)
- COVID-19 Case Analysis Using Power BI | [Link](#)

EDUCATION

Techno Main Salt Lake (B.Tech in Computer Science Engineering)

Kolkata, India

CGPA: 9.24

2020 - 2024

CERTIFICATIONS

- Computation using Python by IIT Madras - **NPTEL**
- Google Cloud Ready Facilitator Program 2022

HONORS & AWARDS

ACM India East and North East Bachelor Dissertation Award 2024

May 2024

- Led a 4 member team for ML based bioinformatics research with strong collaboration and problem solving.
- Awarded 3rd Prize among 36 finalists from over 200 engineering colleges.

ACHIEVEMENTS

- Secured AIR 488 in GATE 2024 (Data Analytics & Artificial Intelligence).
- Completed Hacktoberfest 2021 and 2022 through contributions to 10+ Open Source repositories each year.